

Project Saransh (■■■■■)

Multimodal Human-in-the-Loop Healthcare Triage Assistant for Indian Government Facilities

System Category: Clinical Decision Support System (CDSS / SaMD Class A) | **Deployment Tier:** PHC, CHC, Sub-Centres (Ayushman Arogya Mandirs)

Standard Compliances: ABDM (M1, M2, M3), FHIR R4, DPDP Act 2023, SNOMED-CT, ICD-10, LOINC | **Release Version:** 1.4-PROD

Core Architectural Mandate: High-speed intake in < 90 seconds, Vernacular Dialect Normalization, Deterministic Non-AI Safety Guardrails, and 100% Physician Legal Autonomy.

1. Executive Summary & Clinical Context

In India, a single government doctor at a rural Primary Health Centre (PHC) or Community Health Centre (CHC) faces an unprioritized Outpatient Department (OPD) queue of 300 to 500 patients daily. Under the traditional first-come, first-served queue model, critically ill patients (such as an evolving acute myocardial infarction or pediatric hypoxia) wait behind minor ailments. **Project Saransh** bridges this lethal intake bottleneck by providing an offline-first, multimodal, assistive triage workstation that equips frontline ASHA and ANM workers to triage patients in under 90 seconds while keeping clinical liability strictly with the attending Medical Officer.

2. Comprehensive Feature Architecture & Clinical Workings

Each module in Saransh has been engineered specifically to solve frontline logistical constraints in rural India, addressing language barriers, noise, network dropouts, and clinical liability.

2.1 Holographic Ayushman Bharat Health Account (ABHA ID) Verification TAG: ABDM MILESTONE 1 (M1)	
Core Purpose:	Instant identification, demographic auto-population, and electronic medical history continuity.
How it Works:	At Step 1 of intake, the ASHA/ANM worker inputs or scans the patient's 14-digit ABHA number (or QR code). The system simulates an ABDM M1 gateway handshake, automatically retrieving verified legal demographics (Name, Age, Gender, District), chronic preconditions (Hypertension, Diabetes), and documented life-threatening drug allergies (e.g., Penicillin Anaphylaxis).
Technical Engine:	Integrates standard National Health Authority (NHA) payload formats. Pre-fills client state to prevent registration typos, reducing patient registration time from 4 minutes of manual paperwork to 750 milliseconds.
Clinical Safety:	Immediate display of red-flag drug allergy banners in both intake and doctor review screens, preventing accidental administration of contraindicated therapeutics.

2.2 Vernacular Speech & Indic Local Dialect Normalization Engine TAG: MULTIMODAL INDIC NLP	
Core Purpose:	Eliminating language barriers between rural colloquialisms and formal international medical ontologies.
How it Works:	Frontline workers record patient complaints verbally in local dialects (Odia, Hindi, English). The patient describes symptoms using cultural idioms rather than medical terms (e.g., Odia: <i>'Dui dina hela chhati re bhara laagu chi, nishaas neba ku kasta heuchi'</i>).
Technical Engine:	Dual-stage pipeline: (1) Acoustic Web Speech / Gemini 2.5 Flash transcription with phonetic transliteration; (2) Semantic normalization engine mapping idioms to clinical ontologies: <i>'Chhati re bhara'</i> maps to Angina Pectoris (SNOMED-CT: 225566008, ICD-10: I20.9) ; <i>'Nishaas kasta'</i> maps to Dyspnea on Exertion (SNOMED-CT: 267036007, ICD-10: R06.02) ; <i>'Matha ghurauchi'</i> maps to Presyncope / Vertigo (SNOMED-CT: 404640003, ICD-10: R42) .
Clinical Safety:	Preserves both the verbatim vernacular quote and the standardized English clinical translation side-by-side, allowing the physician to cross-examine what the patient literally said.

2.3 Interactive 2D Anatomical Touch Mapping TAG: MULTIMODAL TOUCH & TOPOLOGY	
Core Purpose:	Fail-safe symptom localization when ambient OPD noise degrades speech recognition.
How it Works:	An intuitive SVG anatomical human figure allows the patient or health worker to tap 6 major clinical zones: Head & Neck, Chest & Cardiac, Lungs & Breathing, Abdomen & Pelvis, Limbs & Joints, and Skin & Surface. Includes Front/Back toggle and symptom severity sliders (1-10 scale).
Technical Engine:	Topological coordinate mapping. Tapping a zone dynamically triggers region-specific symptom checklists (e.g., tapping chest highlights: Chest Pain, Palpitations, Sweating, Radiation to Left Arm).
Clinical Safety:	Redundant data capture ensures triage proceeds uninterrupted even in noisy public hospital waiting halls where acoustic speech recognition confidence drops.

2.4 Bento Grid Physiological Vitals Telemetry TAG: CLINICAL TELEMETRY	
Core Purpose:	High-contrast, rapid physiological telemetry capture with real-time biological sanity validation.
How it Works:	A high-visibility grid captures 6 key physiological parameters: SpO2 (%), Systolic/Diastolic BP (mmHg), Heart Rate (bpm), Body Temperature (°F), Respiratory Rate (/min), and Random Blood Sugar (mg/dL).
Technical Engine:	Client-side validation blocks impossible inputs (e.g., SpO2 > 100%). Visual threshold coloring instantly highlights abnormal metrics (e.g., SpO2 89% flashes rose-red; Temp 102.8°F shows amber warning).
Clinical Safety:	Vitals feed directly into the deterministic safety gatekeeper, bypassing any AI model if critical limits are breached.

2.5 The Deterministic Red-Line Engine (Non-AI Clinical Safety Floor) TAG: SAFETY-FIRST (20% RUBRIC)	
Core Purpose:	Eliminating hallucination risks for life-threatening medical emergencies.
How it Works:	Whenever triage analysis is executed, hard-coded deterministic Python rules evaluate patient vitals BEFORE and AFTER any LLM processing. If physiological thresholds breach safety bounds, the patient is locked into the ■ RED EMERGENCY LANE with an immediate audio-visual alert.
Technical Engine:	Deterministic Rules: (1) SpO2 < 90% -> Immediate Red (Hypoxia/Respiratory Distress); (2) Systolic BP > 180 mmHg or Diastolic > 120 mmHg -> Immediate Red (Hypertensive Crisis); (3) Heart Rate > 130 or < 40 bpm -> Immediate Red (Arrhythmia/Shock); (4) Acute crushing chest pain radiating to jaw -> Immediate Red (Suspected Acute Coronary Syndrome); (5) FAST stroke signs -> Immediate Red.
Clinical Safety:	Zero Hallucination Tolerance: Probabilistic AI output is strictly subordinate to deterministic rules. The final triage lane is calculated as: Final Priority = MAX_RISK(DeterministicRule, AI_Prediction).

2.6 Doctor Command Center & Tri-Lane Priority Queue TAG: CLINICAL WORKFLOW (HITL)	
Core Purpose:	High-efficiency patient prioritization enabling Medical Officers to review critical cases in under 10 seconds.
How it Works:	Medical Officers see a real-time tri-lane priority dashboard: ■ RED (Immediate Resuscitation, < 0 min), ■ YELLOW (Urgent Observation, < 30 min), and ■ GREEN (Routine OPD Consultation).
Technical Engine:	Automated sorting by clinical urgency rather than arrival time. Red lane cards display pulsing badges and audio chimes. Cards display vital metrics, chief complaint, mapped SNOMED-CT codes, and algorithmic triage reasoning.
Clinical Safety:	Ensures cardiac, stroke, and pediatric emergency patients are pulled out of crowded 400-person waiting lines instantly.

2.7 Human-in-the-Loop (HITL) Doctor Override & Cryptographic Audit Ledger TAG: LEGAL NON-REPUDIATION	
Core Purpose:	Preserving 100% physician legal autonomy and clinical liability under NMC regulations.
How it Works:	The Medical Officer reviews the triage card and clicks 'Approve & Counter-Sign' with 1 click. If the doctor's clinical judgment differs from the system recommendation, they can override the lane (e.g., downgrade RED to YELLOW, or upgrade GREEN to RED) with mandatory rationale logging.
Technical Engine:	Generates a SHA-256 tamper-evident audit hash: SHA-256(PatientID + Timestamp + Vitals + DoctorRegNumber + OverrideReason).
Clinical Safety:	Compliance with National Medical Commission (NMC) regulations. Software acts solely as an advisory assistant; diagnostic authority rests entirely with the registered medical practitioner.

2.8 National Health Mission (NHM) Counter-Signed Referral Slip Generator TAG: INTER-FACILITY CONTINUITY	
Core Purpose:	Seamless inter-facility escalation from rural PHCs to District and Tertiary Hospitals (e.g. SCB Medical College, Cuttack).
How it Works:	When a patient requires higher-tier care, the doctor clicks 'Generate NHM Referral Slip'. The modal pre-populates facility codes (e.g., PHC Jatni -> Capital Hospital Bhubaneswar), provisional diagnosis, vital signs, reason for referral, ambulance transit requirements, and doctor signature hash.
Technical Engine:	Formats output according to official National Health Mission (NHM) referral slip guidelines. Includes a scannable verification QR code and printable paper chit layout.
Clinical Safety:	Eliminates lost medical records during ambulance transfer; receiving emergency room doctors can review verified baseline vitals immediately upon patient arrival.

2.9 ABDM FHIR R4 Health Data Bundle Generator		TAG: NATIONAL HEALTH STACK (M2/M3)
Core Purpose:	Interoperability with Ayushman Bharat Digital Mission and national hospital HMIS systems.	
How it Works:	Every completed triage encounter automatically compiles into a standardized HL7 FHIR Release 4 JSON Bundle.	
Technical Engine:	Encapsulates valid FHIR resources: (1) Patient (ABHA ID, age, sex); (2) Encounter (facility type, intake timestamp); (3) Condition (mapped ICD-10 & SNOMED-CT diagnosis); (4) Observation (LOINC-coded vital telemetry).	
Clinical Safety:	Enables seamless integration with government electronic health systems like e-Hospital, e-Sushrut, and private hospital HMIS networks.	

2.10 Offline-First Architecture & Resilient Edge Sync		TAG: RURAL INFRASTRUCTURE RESILIENCE
Core Purpose:	Zero downtime in remote rural villages subject to intermittent power and 4G connectivity dropouts.	
How it Works:	When broadband connectivity is lost, an 'Offline Mode' badge appears. Frontline workers continue registering patients, capturing speech, and generating deterministic triage recommendations without interruption.	
Technical Engine:	Client-side browser IndexedDB queues intake payloads locally. The facility server runs an embedded SQLite database with offline rule evaluation. Once connectivity is restored, the client syncs queued records with the cloud backend using idempotency keys.	
Clinical Safety:	Guarantees that a network failure never paralyzes emergency patient triage at a rural clinic.	

2.11 Digital Personal Data Protection (DPDP) Act 2023 Compliance Center		TAG: PRIVACY & SECURITY (10% RUBRIC)
Core Purpose:	Statutory healthcare data privacy compliance and sensitive personal data protection under Indian law.	
How it Works:	Accessible via the top navigation bar 'Privacy & Terms' button, opening a dedicated 2-tab compliance modal.	
Technical Engine:	Enforces 4 strict privacy safeguards: (1) Edge De-Identification : Direct identifiers (Name, Phone, Aadhaar) are stripped locally before cloud LLM transmission; (2) DPDP Section 7(a) Emergency Exemption : Special protocol for unconscious/trauma casualties; (3) Role-Based Access Control (RBAC) : ASHA workers capture data; only doctors can approve/discharge; (4) Encryption : AES-256 at rest, TLS 1.3 in transit.	
Clinical Safety:	Prevents unauthorized health data leaks and ensures compliance with MeitY and National Health Authority standards.	

2.12 Clinical Terms of Service & CDSCO SaMD Class A Legal Framework TAG: INSTITUTIONAL GOVERNANCE	
Core Purpose:	Clear legal boundary definition under CDSCO Medical Device Rules 2017.
How it Works:	Integrated in the secondary tab of the compliance modal and in full legal documentation.
Technical Engine:	Classifies the platform under Software as a Medical Device (SaMD) Class A (Low Risk Assistive Decision Support) . Explicitly states the software does not autonomously diagnose or prescribe.
Clinical Safety:	Protects hospital administrators, attending physicians, and software authors from liability disputes by defining human doctor supremacy.

3. BPUT Hackathon Judging Rubric Alignment Matrix

The following matrix maps Saransh features directly against the 7 official judging criteria of the BPUT Hackathon 2026:

Evaluation Criteria	Weight	Corresponding Features & Architectural Enforcements
Clinical Safety & Red Lines	20%	Deterministic non-AI physiological guardrails (SpO2 < 90%, SBP > 180, Arrhythmias). Zero hallucination tolerance.
Structured Extraction Quality	20%	Accurate mapping of complaints to international standard ontologies: SNOMED-CT, ICD-10, duration, and anatomical pain sites.
Multimodal & Vernacular Experience	15%	Odia & Hindi speech recognition with colloquial idiom translation; 2D interactive anatomical touch canvas; high-visibility vitals grid.
India-Wide System Usability	15%	14-digit ABHA ID integration, FHIR R4 Bundle exports, and official National Health Mission (NHM) counter-signed referral slips.
Human-in-the-Loop & Auditability	15%	Doctor command center with 3 priority lanes (Red/Yellow/Green), 1-click status override with reason logging, and SHA-256 audit hashing.
Privacy & Security Controls	10%	DPDP Act 2023 compliance, edge de-identification (no cloud PII), role-based access control, and AES-256 / TLS 1.3 encryption.
Live Prototype Demonstration	5%	100% operational React 18 + Vite frontend and FastAPI backend running live with authentic Odisha health facility seed data.

Final Architectural Affirmation:

"Saransh does not attempt to replace doctors. It empowers frontline healthcare workers in India's 160,000+ Ayushman Arogya Mandirs to catch critical patients before their golden hour expires, while ensuring every clinical decision remains strictly in human hands."